

①

$$x(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\sin\left(\frac{5\pi}{3}t\right) \rightarrow \text{signal periódico no tempo}$$

a) ??

Como $\frac{2\pi}{3} \neq \frac{5\pi}{3}$, então temos que encontrar o T mínimo entre estes 2; logo vamos ter que:

$$\omega_0 = \frac{\pi}{3}$$

de forma a que multiplicar por um $m \in \mathbb{Z}$ dê estes 2 T !!

b)

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}, \quad k = 0, \pm 1, \pm 2, \dots$$

$$\omega_0 = \pi/3 \Rightarrow \frac{2\pi}{T} = \frac{\pi}{3} \Rightarrow T = 6$$

vai ser dado por: $a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$

mas neste caso, m não há necessidade de se ter aqui pois temos que:

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad \text{e} \quad \sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Substituindo, vamos ter que:

$$\begin{aligned} x(t) &= 2 + \frac{1}{2} e^{j2\omega_0 t} + \frac{1}{2} e^{-j2\omega_0 t} + \frac{2}{j} e^{j5\omega_0 t} - \frac{2}{j} e^{-j5\omega_0 t} \\ &= 2 + \frac{1}{2} e^{j2\omega_0 t} + \frac{1}{2} e^{-j2\omega_0 t} - 2j e^{j5\omega_0 t} + 2j e^{-j5\omega_0 t} \end{aligned}$$

$$a_0 = 2$$

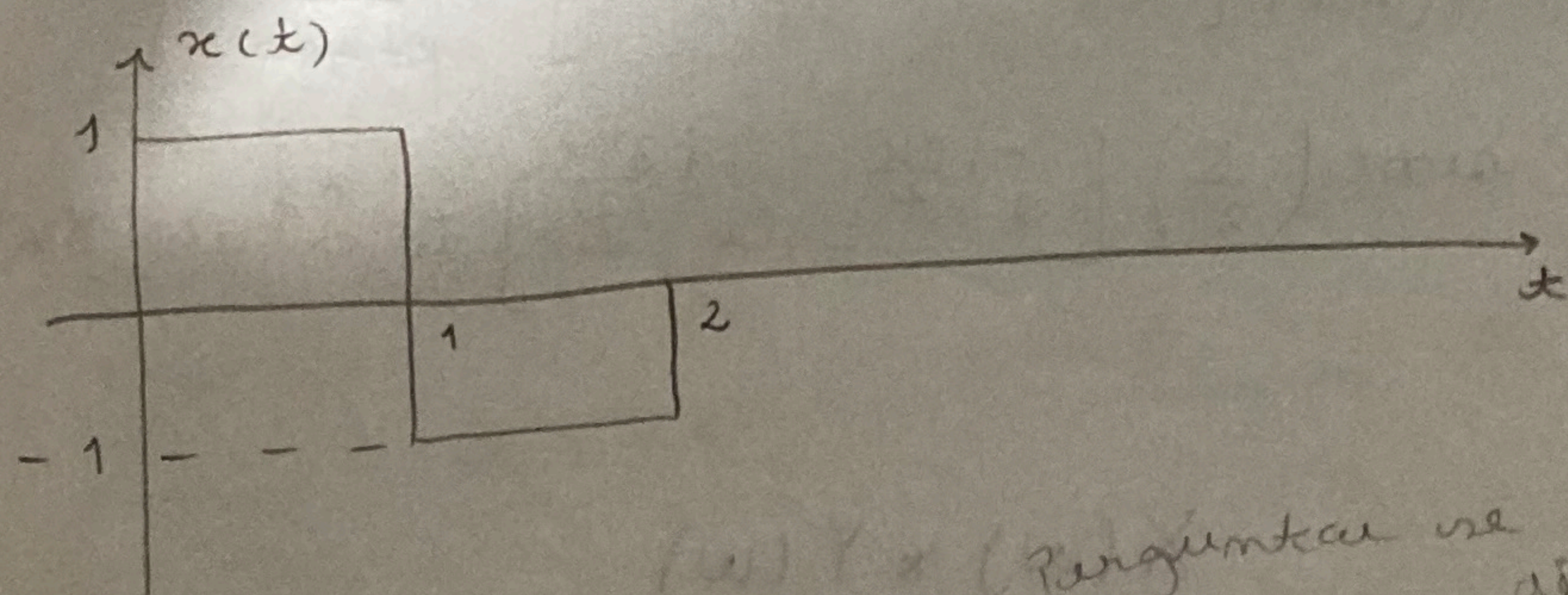
$$a_5 = -2j$$

$$a_2 = \frac{1}{2}$$

$$a_{-5} = 2j$$

$$a_{-2} = \frac{1}{2}$$

②

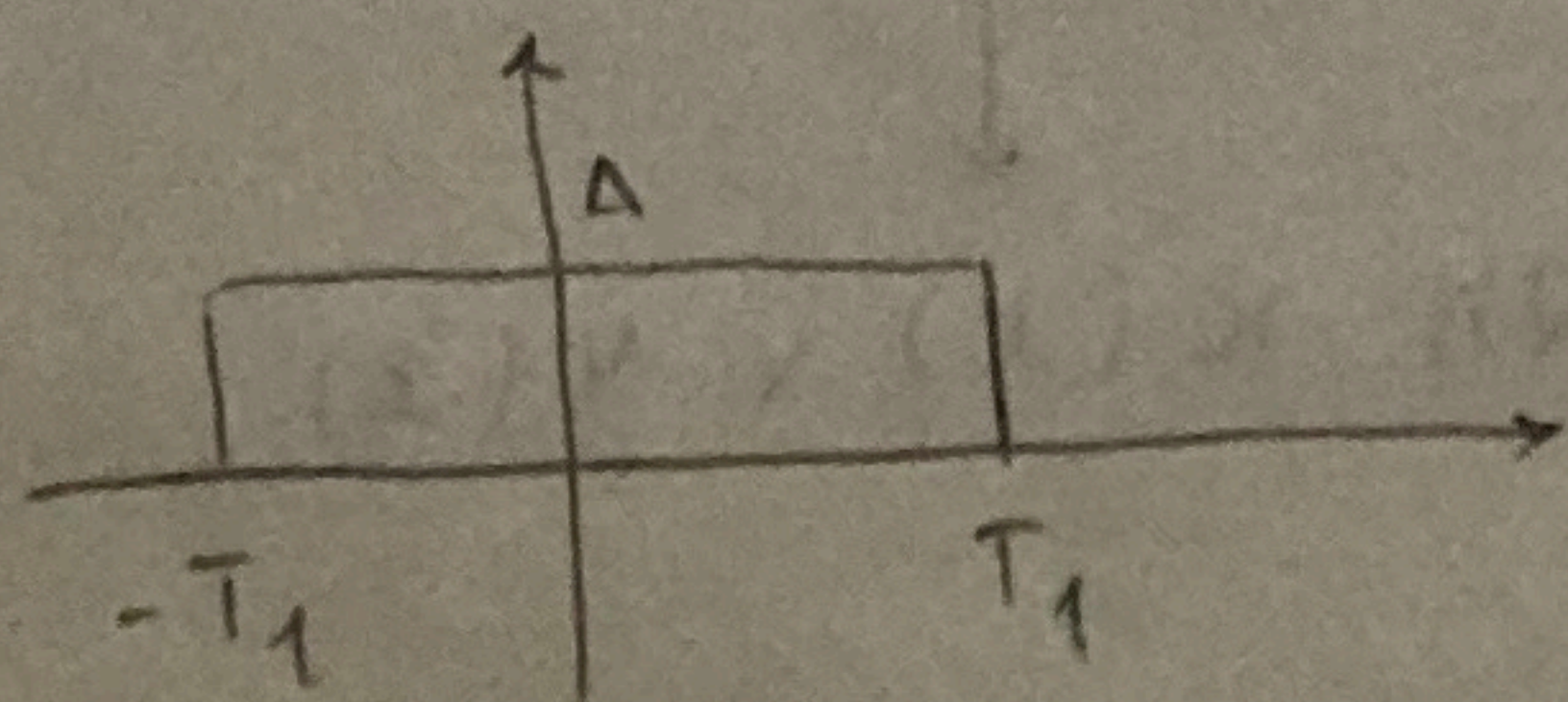


signal m periódico no tempo

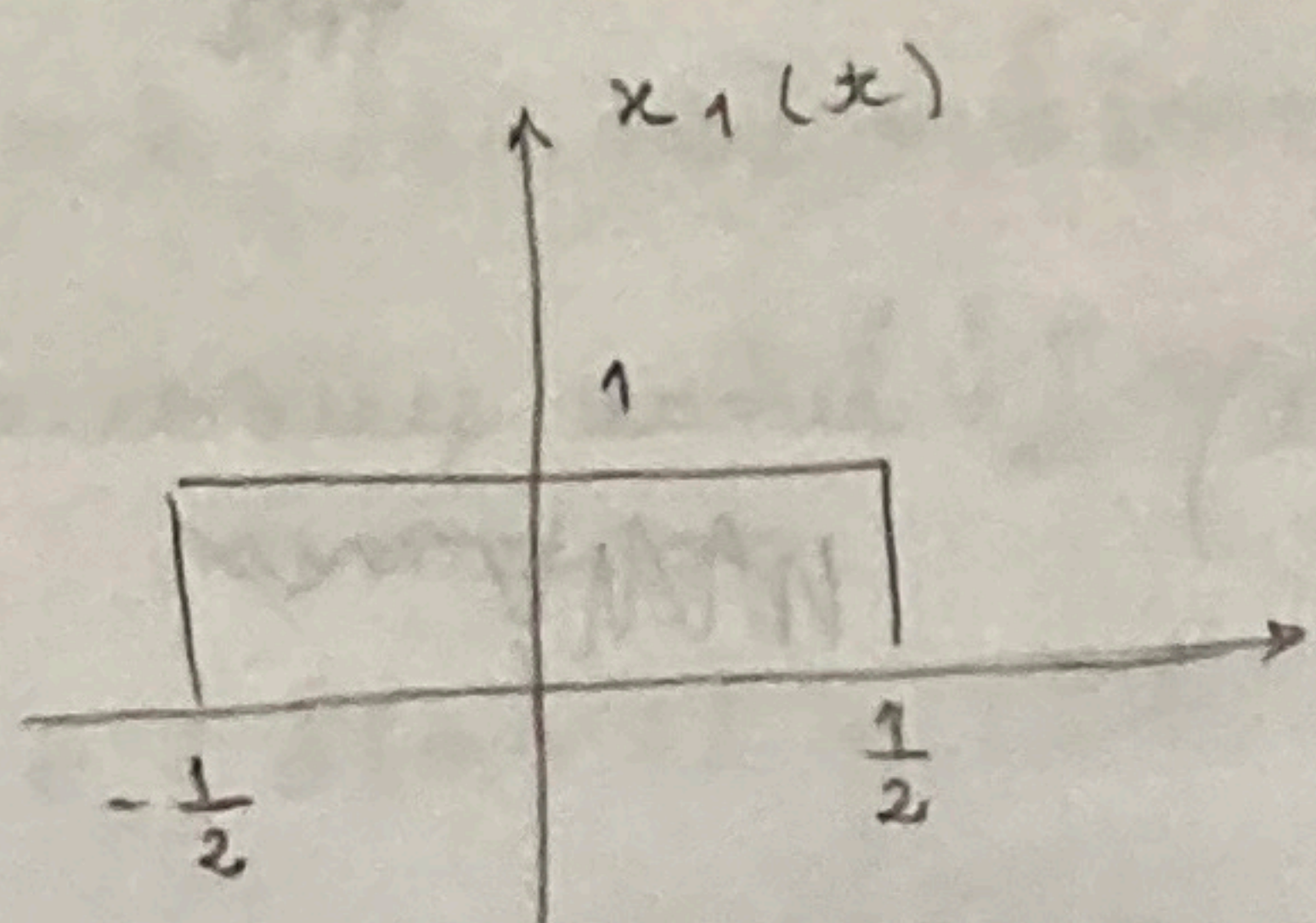
Sabemos que:

a)

$$X(\omega) = ??$$



Transformada de Fourier $\rightarrow 2AT_1 \operatorname{sinc}\left(\frac{\omega T_1}{\pi}\right)$



Pela propriedade da linearidade:
 $x(t) = x_1(t + \frac{1}{2}) - x_1(t - \frac{1}{2})$

Logo, $X(\omega) = X_1(\omega) e^{-j\omega \frac{1}{2}} - X_1(\omega) e^{-j\omega \frac{3}{2}}$

$$X_1(\omega) = 2 \times 1 \times \frac{1}{2} \text{sinc}\left(\frac{\omega \times \frac{1}{2}}{\pi}\right) = \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$X(\omega) = \text{sinc}\left(\frac{\omega}{2\pi}\right) \left[e^{-j\omega \frac{1}{2}} - e^{-j\omega \frac{3}{2}} \right]$$

$$g(t) = \sum_{k=-\infty}^{+\infty} x(t - kT), \quad T = 200$$

$$\Rightarrow g(t) = \sum_{k=-\infty}^{+\infty} x(t - 2k) = x(t) * \sum_{k=-\infty}^{+\infty} \delta(t - 2k)$$

Pela propriedade da convolução:

$$x(t) * y(t) \xrightarrow{\text{CTFT}} X(\omega) Y(\omega)$$

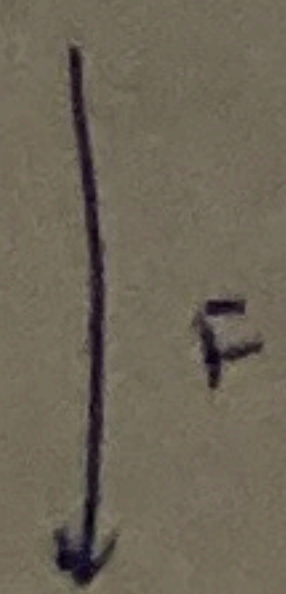
Sabemos o $X(\omega)$ da alínea anterior e $Y(\omega)$ vai ser dado por:

$$\sum_{m=-\infty}^{+\infty} \delta(t - mT) \xrightarrow{\text{CTFT}} \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$$

Logo vamos ter que:

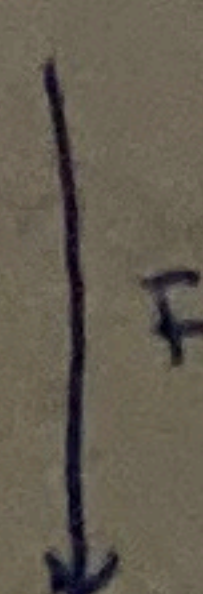
$$\begin{aligned} G(\omega) &= X(\omega) Y(\omega) = \\ &= \text{sinc}\left(\frac{\omega}{2\pi}\right) \left[e^{-j\omega \frac{1}{2}} - e^{-j\omega \frac{3}{2}} \right] \pi \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right) = \\ &= \sum_{k=-\infty}^{+\infty} \text{sinc}\left(\frac{k}{2}\right) \left[e^{-j\frac{\pi k}{2}} - e^{-j\frac{3\pi k}{2}} \right] \pi \delta\left(\omega - \frac{2\pi k}{T}\right) \end{aligned}$$

$$x(t) * y(t)$$



$$X(\omega) \times Y(\omega)$$

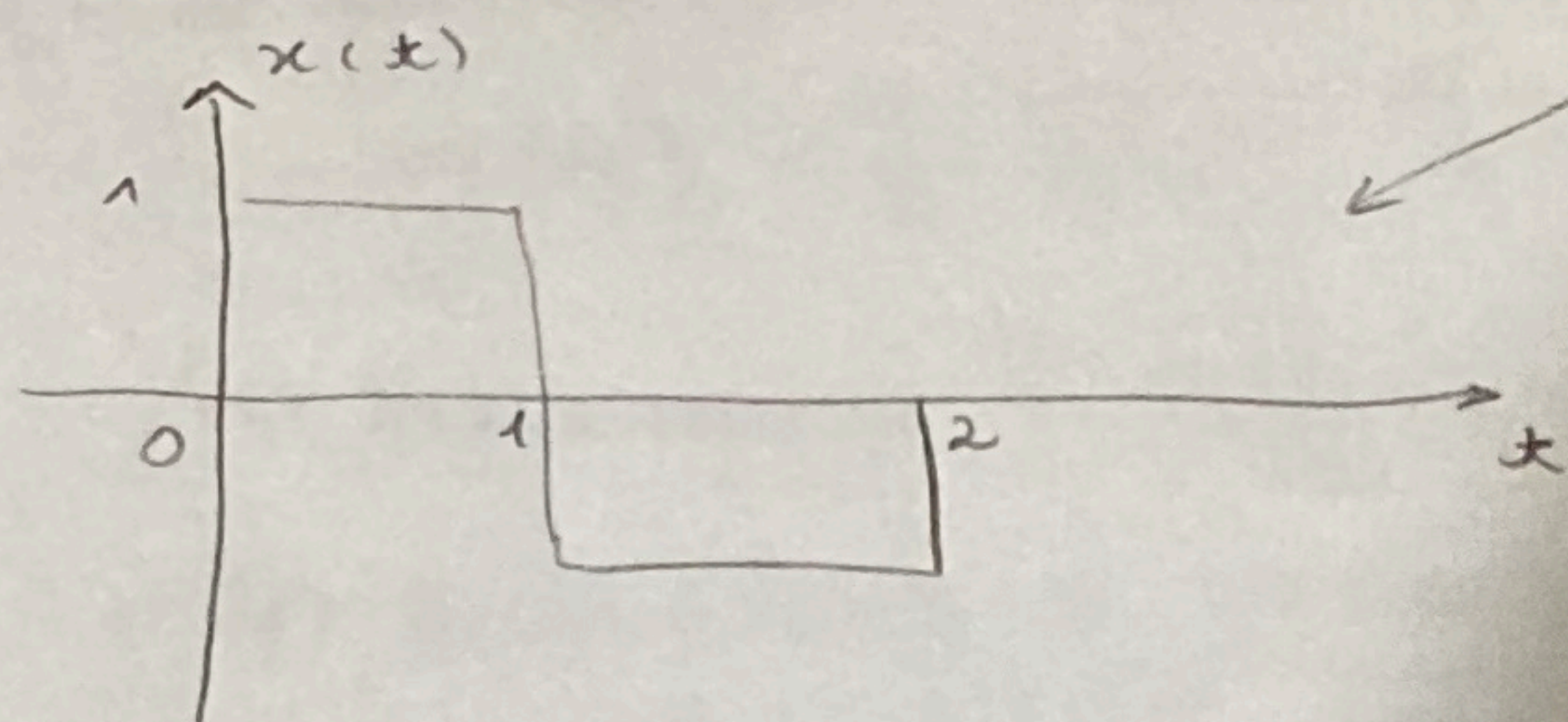
$$X(\omega) * Y(\omega)$$



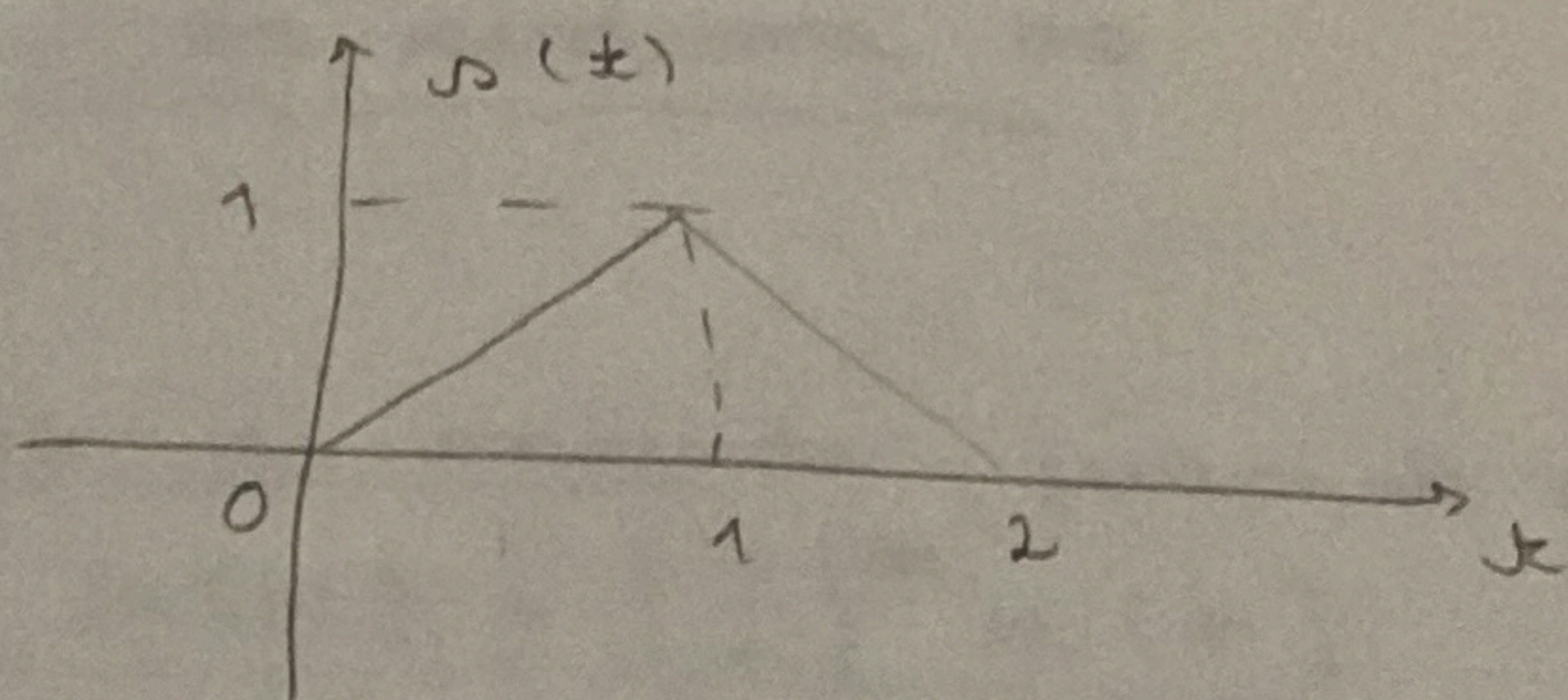
$$2\pi x(t) \times y(t)$$

~~xxxx~~
 e) ver pela fórmula!!
 $s(t) = \int x(t) dt$

Sabemos que $x(t)$ é dado por:



o integral deste sinal
 é igual a $s(t)$.
 Logo, vamos
 ter que:



Sabemos que:

$$\frac{d}{dt} s(t) \xrightarrow{F} j\omega S(\omega) = X(\omega)$$

Logo, $X(\omega) = j\omega S(\omega) \Leftrightarrow \frac{X(\omega)}{j\omega} = S(\omega)$

Por isso, vamos ter que:

$$S(\omega) = \frac{\text{sinc}\left(\frac{\omega}{2\pi}\right)}{j\omega} \left[e^{-j\frac{\omega}{2}} - e^{-j\frac{3\omega}{2}} \right]$$

d)
 $x(t) = \sum_{k=-\infty}^{+\infty} s(t - kT)$, $T = 2s$ $R(\omega) = ??$

$$\downarrow$$

$$x(t) = \sum_{k=-\infty}^{+\infty} s(t - 2k)$$

$$= s(t) * \sum_{k=-\infty}^{+\infty} \delta(t - 2k)$$

Na propriedade da convolução:

$$x(t) * y(t) \xrightarrow{\text{CTFT}} X(\omega) Y(\omega)$$

$S(\omega) \rightarrow$ da última anterior

$$\sum_{k=-\infty}^{+\infty} \delta(t - 2k) \xrightarrow{\text{CTFT}} \pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \pi k)$$

$$R(\omega) = S(\omega) \pi \sum_{k=-\infty}^{+\infty} \delta(\omega - \pi k)$$

(continuar os cálculos!!)